

MUTSU, Japan

WMO: 475760

Lat: 41.2833N

Lon: 141.2114E

Elev: 16

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	20.2	8.8	8.6	22.0	11.3	9.8	24.2	20.4	32.9	18.2	32.3	4.3	290	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	84.5	75.2	81.7	73.6	79.1	72.0	76.5	82.3	75.1	80.1	73.6	77.9	8.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	129.7	80.1	73.4	124.1	78.7	72.0	118.5	77.1	40.0	82.6	38.7	80.4	37.3	78.2	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	15.9	13.7	10.2	89.1	4.3	2.7	7.2	91.0	4.6	92.6	2.2	94.1	-0.9	96.1		
			DB	WB	9.7	78.5	4.3	1.5	6.6	79.6	4.1	80.5	1.7	81.3	-1.4	82.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	29.0	29.9	35.7	45.8	54.9	61.3	68.9	72.0	65.8	54.5	43.6	33.8
	DBStd	15.76	4.87	5.16	5.48	5.85	5.39	5.03	5.59	5.02	5.28	5.48	6.35	5.31
	HDD50	2557	650	564	445	153	17	0	0	0	0	19	207	502
	HDD65	6049	1115	984	909	577	314	131	27	5	52	328	641	966
	CDD50	2453	0	0	0	26	170	340	586	682	475	158	16	0
	CDD65	472	0	0	0	0	2	21	148	222	77	2	0	0
	CDH74	2248	0	0	0	1	19	74	671	1185	296	2	0	0
	CDH80	422	0	0	0	0	1	3	121	257	40	0	0	0
Wind	WSAvg	6.0	6.1	6.3	6.9	6.9	6.4	5.5	5.2	5.0	5.1	5.5	6.3	6.3
Precipitation	PrecAvg	52.1	3.9	3.1	3.0	3.0	3.7	3.5	5.2	6.1	6.6	4.4	4.8	4.3
	PrecMax	65.8	8.2	5.9	4.6	6.5	6.6	9.5	14.2	11.9	12.6	8.7	8.9	7.2
	PrecMin	38.5	1.8	1.3	1.4	0.4	1.2	0.7	0.9	1.9	1.5	1.7	2.2	1.5
	PrecStd	6.2	1.5	1.3	0.9	1.3	1.4	1.9	2.7	2.6	2.8	1.9	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	46.3	57.3	69.5	76.6	79.7	86.9	89.0	84.2	72.5	64.0	52.9
		MCWB	40.2	41.1	47.7	54.0	60.5	66.9	75.8	77.2	74.1	63.6	56.7	48.8
	2%	DB	41.0	42.8	52.0	63.6	72.1	76.0	83.5	85.5	80.3	69.3	60.2	48.8
		MCWB	37.0	38.1	44.6	51.7	58.6	65.6	74.4	76.1	72.4	61.6	54.1	44.6
	5%	DB	38.6	40.6	48.1	59.3	68.5	73.3	80.5	82.9	77.2	67.1	57.6	45.8
		MCWB	34.9	36.1	42.1	49.3	57.1	64.5	72.9	74.8	69.9	60.2	52.2	42.0
	10%	DB	36.8	38.4	44.9	55.9	64.9	70.6	77.7	80.2	74.8	64.9	55.0	42.9
		MCWB	33.1	34.3	39.6	47.6	56.0	63.4	71.6	73.5	68.3	58.9	50.3	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.2	42.8	48.8	56.5	63.6	69.9	77.2	78.9	76.3	66.8	58.5	49.7
		MCDB	43.2	45.1	54.4	64.7	71.9	75.6	84.5	86.5	81.7	69.8	61.6	52.7
	2%	WB	37.7	38.9	45.7	53.5	60.9	67.7	75.5	77.2	74.0	63.7	55.6	45.8
		MCDB	40.1	42.0	50.8	60.1	68.1	73.6	81.4	83.3	78.5	67.4	59.2	48.1
	5%	WB	35.5	36.8	42.9	51.0	58.9	66.0	74.1	76.0	71.8	61.8	53.3	42.3
		MCDB	38.1	39.8	47.3	57.3	66.2	71.5	79.1	81.2	75.7	65.7	56.6	45.4
	10%	WB	33.7	34.7	40.4	48.7	56.9	64.2	72.5	74.6	69.8	60.0	50.9	39.3
		MCDB	36.2	37.7	44.3	54.5	63.3	69.3	76.9	79.0	73.4	63.9	54.4	42.6
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.3	13.1	15.9	15.9	13.9	11.7	11.7	14.4	16.9	14.0	10.8
		MCDBR	12.4	13.9	18.3	22.1	22.9	18.7	16.0	15.6	16.0	18.5	18.9	15.2
		MCWBR	10.4	10.9	12.8	12.5	11.8	10.2	8.3	7.8	9.3	12.4	14.3	12.9
	5% WB	MCDBR	12.0	13.7	17.0	19.4	19.8	16.1	14.3	14.0	13.8	15.4	17.2	15.1
		MCWBR	10.8	11.7	13.1	12.6	11.7	9.7	7.9	7.5	8.6	11.5	14.5	13.6
Clear-Sky Solar Irradiance	taub	0.365	0.409	0.481	0.535	0.551	0.524	0.523	0.492	0.434	0.413	0.397	0.380	
	taud	2.116	1.999	1.874	1.804	1.853	2.006	2.082	2.181	2.298	2.219	2.153	2.101	
	Ebn at Noon	237	244	239	237	237	244	243	246	254	242	224	218	
	Edh at Noon	34	45	57	66	64	55	51	45	37	36	33	32	
All-Sky Solar Radiation	RadAvg	452	698	1067	1428	1563	1535	1357	1308	1158	886	533	378	
	RadStd	24	60	71	145	158	123	145	141	102	71	45	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page